Clinical Protocol RAG

Applicability-aware clinical recommendation prototype

Presented by

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Presented to

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Clinical Protocol RAG

Applicability-aware clinical recommendation prototype

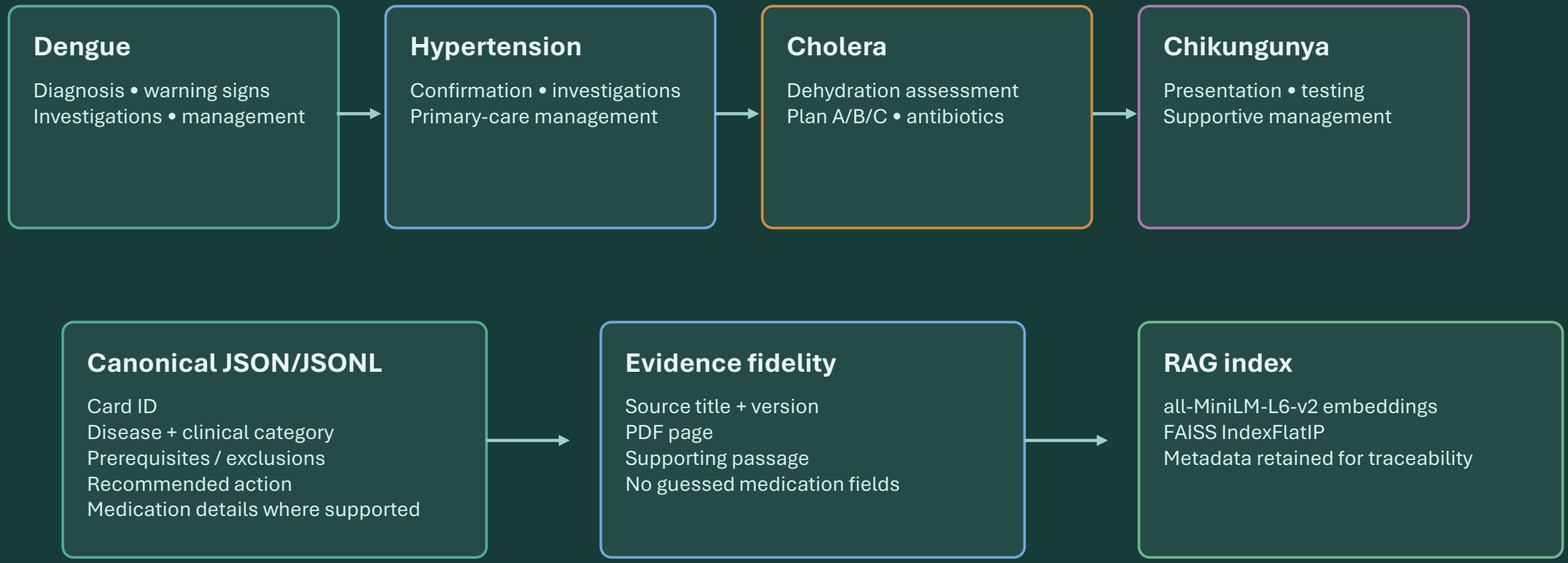
4 clinical protocols | **17 canonical cards** | **2 runnable modes** | **6 frozen cases** | **12 saved outputs**

Goal

Given patient symptoms and available context, retrieve protocol evidence and produce a probable diagnosis, investigations, medication planning, advice, and explicit missing-information handling.

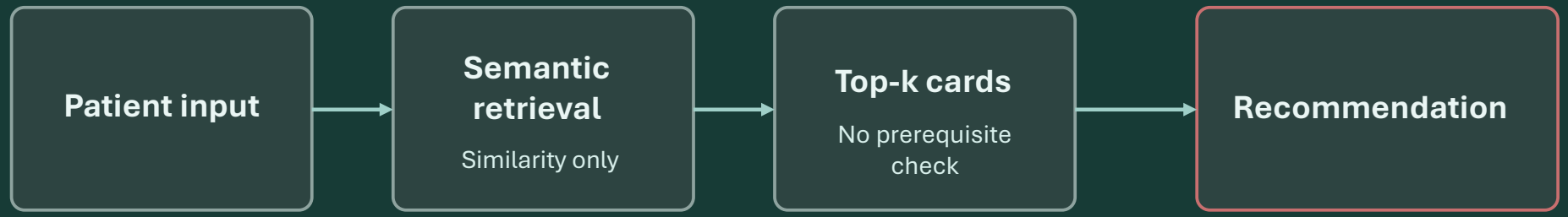
Interview presentation • prototype + experiment + paper discussion

From protocols to retrievable clinical knowledge

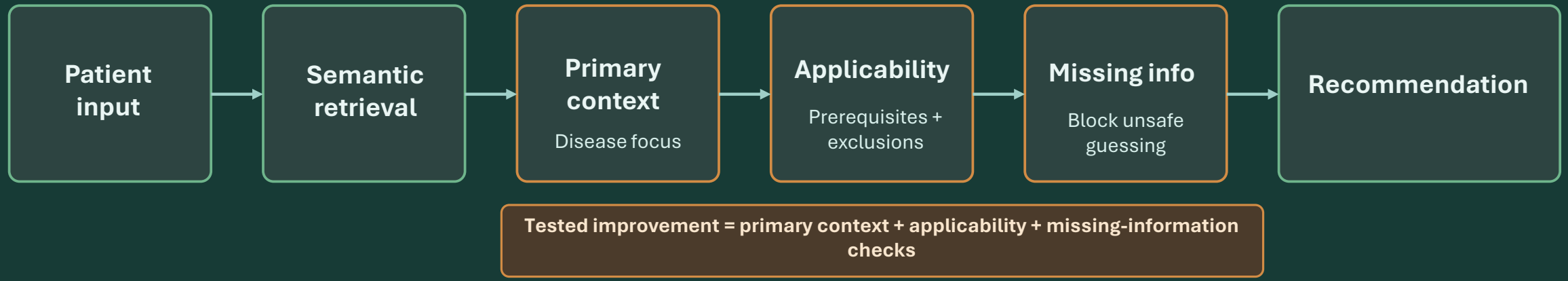


Baseline vs. tested improvement

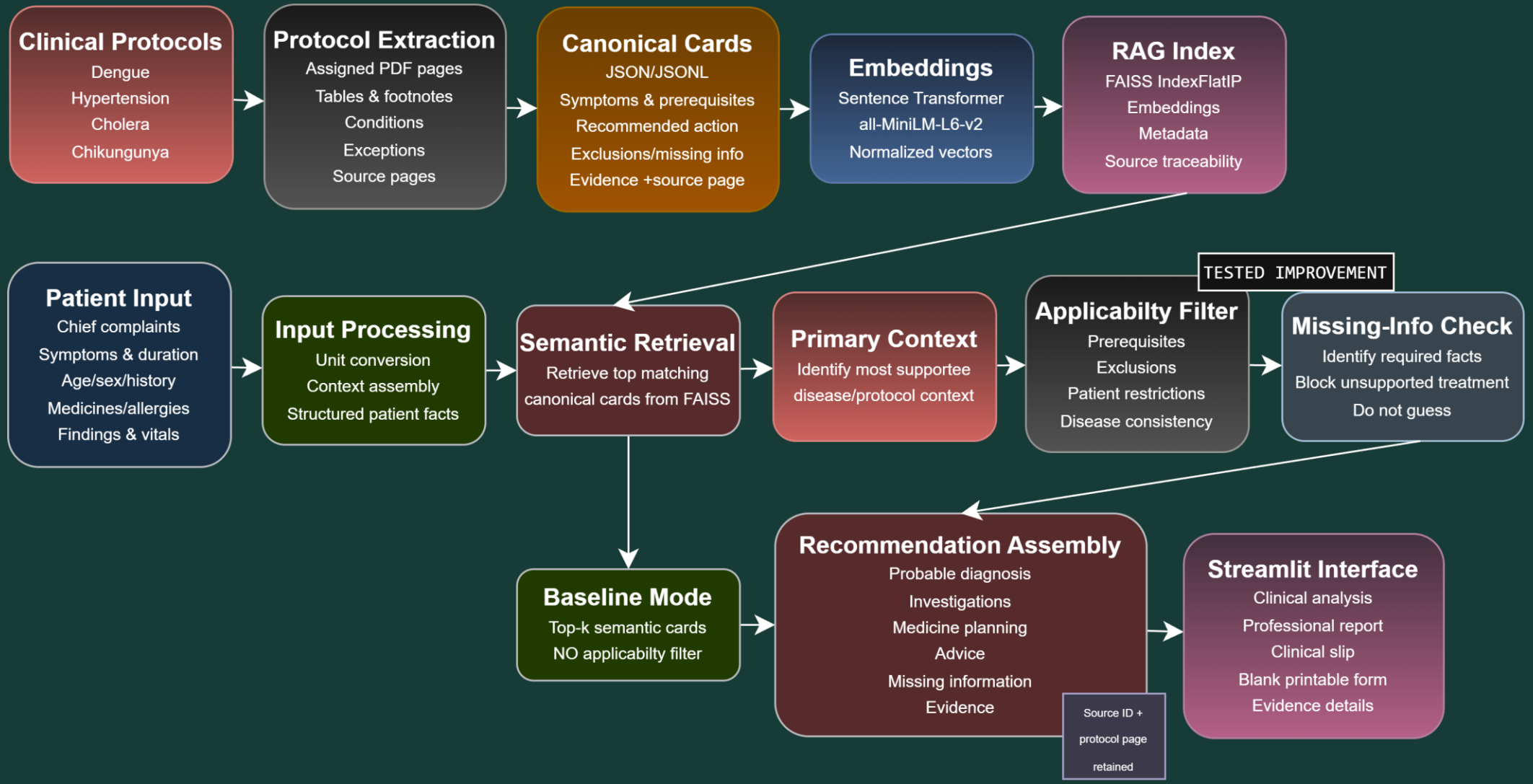
Baseline



Improved



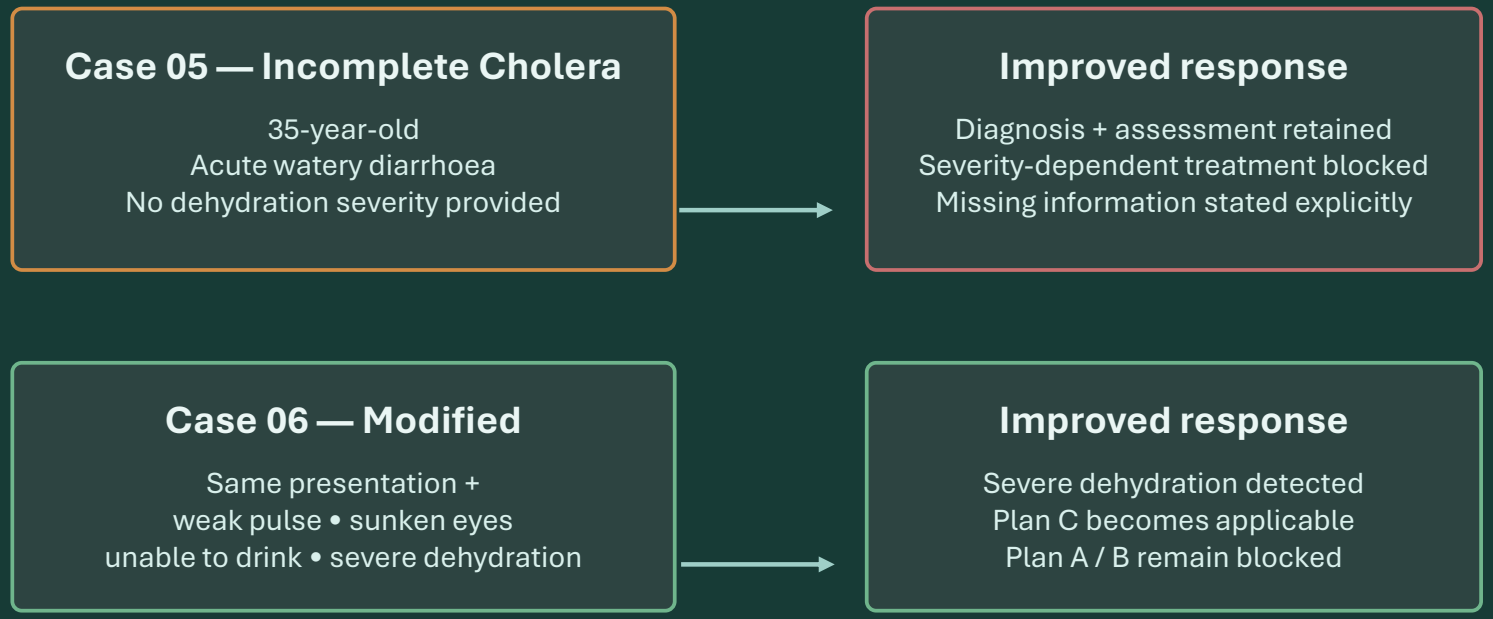
System architecture



Future: NER input • doctor acceptance/feedback • clinician-approved prescription assembly

Demo: why applicability matters

Same disease context, one added prerequisite changes the management pathway.

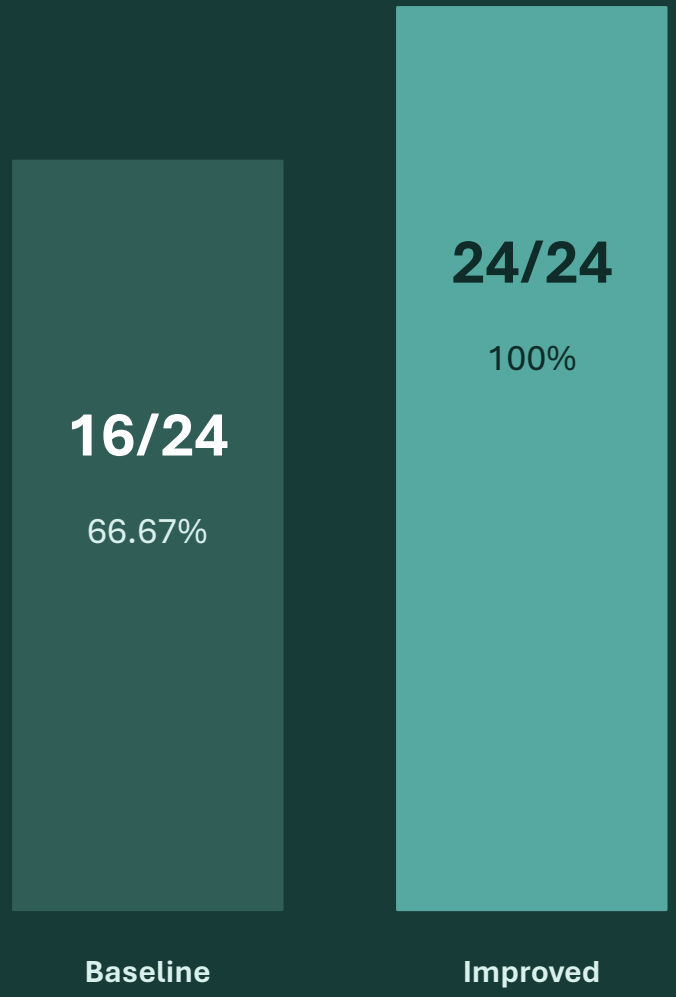


Evaluation on six frozen fictional cases

Four predefined checks per output; 12 outputs total.

Predefined checks

- Expected behavior occurred
- Recommendations supported by citations
- Relevant restrictions preserved
- Missing information handled without guessing



Latency

Baseline avg: 0.0469 s
Improved avg: 0.0279 s

Where baseline failed

Cholera: incompatible treatment plans
Incomplete case: treatment despite missing severity
Chikungunya: cross-disease retrieval leakage

24/24 = predefined safety-and-behavior checks on six self-authored cases — NOT clinical accuracy or independent validation.

Related paper: multimodal AMIE (Saab et al., 2026)

Research problem

Text-only diagnostic dialogue misses images, ECGs, and clinical documents used in real care.

Design

Multimodal AMIE + state-aware dialogue phases that update patient state, DDx, and knowledge gaps.

Evaluation

Randomized, blinded exploratory OSCE-style study with 105 simulated telehealth consultations.

Result

Higher performance on 29/32 assessed axes; 7/9 multimodal-reasoning metrics.

What I adopt

- Maintain an evolving patient state
- Make knowledge gaps explicit
- Use new evidence to refine recommendations

Bangladesh GP transfer limitation

- Simulated telehealth, not real Bangladesh primary care
- Local prevalence, language and resource constraints differ
- Full multimodal infrastructure exceeds this assessment scope

Source: Saab et al., Nature Medicine 32, 1726–1736 (2026) • DOI: 10.1038/s41591-026-04371-0

Takeaways, limitations & next steps

What worked

- Protocol evidence remained traceable
- Applicability rules reduced prerequisite errors
- Missing information was surfaced instead of guessed
- Same UI supports baseline and improved modes

Current limitations

- Only four protocols / 17 cards
- Six self-authored fictional cases
- No real clinician validation
- Rule coverage is intentionally narrow
- Local prototype timing only

Future integration

- NER extraction from notes
- Doctor accept / reject / modify feedback
- Broader protocol coverage
- Blinded clinician review
- Clinician-approved prescription assembly

Core contribution

A small, auditable clinical RAG prototype that separates semantic relevance from patient-specific applicability before producing a recommendation.

Clinical Protocol RAG

Thank You